Complied vs Interpreted

TEAM 2



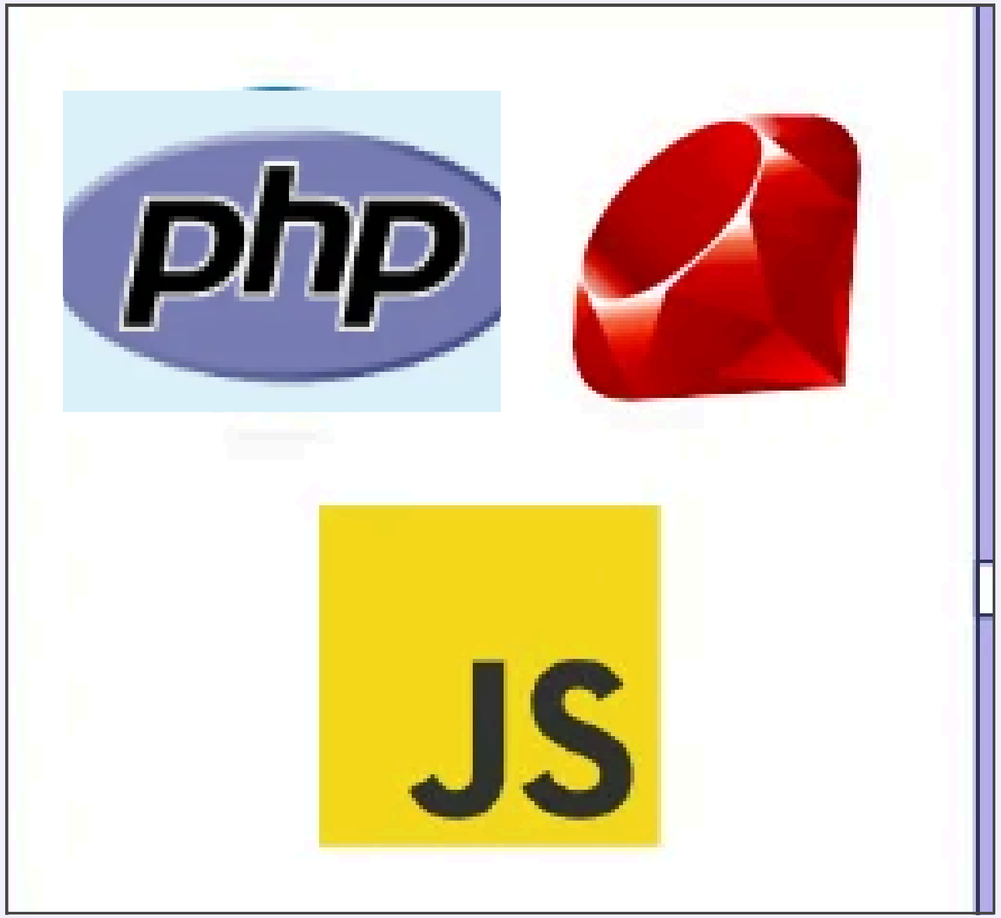
Compiled

- Written specifically to work with the computers
- Doesn’t require any extra processing power
- Executes program in it’s entirety first
- Translated into machine code (Binary)

Compiled Languages include ones like: C, C++, and Swift

Interpreted

- Built for Performance and Scalability
- Traditionally slower due to runtime overhead
- Runtime/line-by-line error reporting
- Translation is done at runtime by an environment tool called an interpreter.



Comparison

Compiled Languages

- Extremely fast to process
- More strict towards errors
- Can be compiled to create an executable program and not show source code

Interpreted Languages

- Slower to process
- More lenient towards errors,
- Easier to debug
- Can’t be compiled or even run without the interpreter

Why Python is Both



- Written code is compiled into bytecode
- Python’s built in interpreter (CPython) will execute that code
- This design is done to balance programmer efficiency, cross-platform compatibility, and runtime performance.

Simplest explanation would be that Python borrows both features from compiled and interpreted code, making it a hybrid.

Thank you